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(a)	First Law of Thermodynamics states that the <u>increase in internal energy of a system</u> is the sum of the <u>heat supplied to</u> the system and the <u>work done on</u> the system.				
(b)	$q_{to} = (2.26 \times 10^6)(5.0)$				
	$\Delta V = \left(\frac{5.0}{0.598} - \frac{5.0}{1000} \right)$				
	$w_{on} = -(1.01 \times 10^5) \left(\frac{5.00}{0.598} - \frac{5.00}{1000} \right)$				
	$\Delta U = q_{to} + w_{on}$ $= 1.05 \times 10^7 \text{ J (3 s.f.)}$				
(c)(i)	As the product of p and V at state C is greater than the product of p and V at state D, <u>OR</u> $\Delta U = q_{to} + w_{on}$ Since q_{to} is 0 and w_{on} is negative due to expansion, State C is at a higher temperature.				
(c)(ii)	section of cycle	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J	
	A to B	0	300	300	
	B to C	2580	- 740	1840	
	C to D	0	- 440	- 440	
	D to A	-1700	0	- 1700	
Q4					